

Maths: Statistics

Normal Distribution & Z-Scores | Grade 6

Learning Objectives

By the end of this lesson, you will be able to:

- Describe the shape and properties of a normal distribution
- Apply the empirical (68-95-99.7) rule
- Calculate Z-scores using the formula $Z = (X - \mu) \div \sigma$
- Interpret positive and negative Z-scores
- Solve real-life problems using Z-scores

1. Key Vocabulary

Term	Definition
Normal Distribution	A bell-shaped, symmetrical pattern where most data clusters near the mean.
Mean (μ)	The average value of a dataset; the centre of a normal distribution.
Standard Deviation (σ)	Measures how spread out the data is around the mean.
Z-Score	Tells us how many standard deviations a value is above or below the mean.
Empirical Rule	The 68-95-99.7 rule for data distribution within standard deviations.
Bell Curve	The symmetrical graph shape of a normal distribution.

2. The Normal Distribution

A normal distribution describes how data is spread around an average value. It forms a symmetrical **bell curve**. Many real-life measurements follow this pattern, including heights, test scores, and shoe sizes.

Properties

- **Symmetrical:** Left and right halves are mirror images.
- **Equality:** Mean = Median = Mode at the exact centre.
- **Area:** The total area under the curve is equal to 1 (100%).
- **Tails:** The curve never touches the x-axis (infinite tails).

The Empirical Rule (68-95-99.7)

Range	Percentage	Meaning
$\pm 1\sigma$	68.3%	Most data falls within 1 SD of the mean.
$\pm 2\sigma$	95.4%	Almost all data falls within 2 SDs of the mean.
$\pm 3\sigma$	99.7%	Nearly all data; only 0.3% is outside this range.

WORKED EXAMPLE

A test has $\mu = 60$ and $\sigma = 8$.

- 68.3% score between 52 and 68 (60 ± 8)
- 95.4% score between 44 and 76 (60 ± 16)

3. Z-Scores

Z-scores allow us to compare results from different datasets by standardizing them.

The Formula

$$Z = (X - \mu) \div \sigma$$

Interpretation Guide

Z-Score	Interpretation
$Z = 0$	The score is exactly the average (mean).
<i>Positive (+)</i>	The score is above average.
<i>Negative (-)</i>	The score is below average.
$Z > +3$ or < -3	An exceptionally unusual value.

EXAMPLE CALCULATION

Scenario: Math Test ($\mu = 75$, $\sigma = 5$). Aisha scored 85.

Calculation: $Z = (85 - 75) \div 5 = 10 \div 5 = 2.0$

Result: Aisha is 2 standard deviations above the mean.